BOUNDED DERIVED CATEGORIES ARE TRIANGULATED SUBCATEGORIES

THE STACKS PROJECT AUTHORS

Editorial note. The D_H^- and D_H^b variants repeat the same argument with a changed boundedness window (mirror variants), transcribed as-is.

Allowed standard prerequisites (editorial).

- Triangulated subcategory criterion. lemma-triangulated-subcategory (same chapter, cited in-span; not carded). Supplies the named-subcategory engine for triangle closure.
- Long exact cohomology sequence. equation-long-exact-cohomology-sequence (same chapter, cited in-span; not carded). Drives the saturation argument.

Source reference. Original equation 0148: sources/stacks-derived/pinned-original-source.tex, source lines 499–499. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05QX: sources/stacks-derived/pinned-original-source.tex, source lines 1077–1077. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $H : \mathcal{D} \to \mathcal{A}$ be a homological functor of a pre-triangulated category into an abelian category. Let $\mathcal{D}_H^+, \mathcal{D}_H^-, \mathcal{D}_H^b$ be the full subcategory of $\mathcal{D}$ with objects*

$$\begin{aligned} \mathrm{Ob}(\mathcal{D}_H^+) &= \{X \in \mathrm{Ob}(\mathcal{D}) \mid H(X[n]) = 0 \text{ for all } n \ll 0\} \\ \mathrm{Ob}(\mathcal{D}_H^-) &= \{X \in \mathrm{Ob}(\mathcal{D}) \mid H(X[n]) = 0 \text{ for all } n \gg 0\} \\ \mathrm{Ob}(\mathcal{D}_H^b) &= \{X \in \mathrm{Ob}(\mathcal{D}) \mid H(X[n]) = 0 \text{ for all } |n| \gg 0\} \end{aligned}$$

Each of these is a strictly full saturated pre-triangulated subcategory of $\mathcal{D}$. If $\mathcal{D}$ is a triangulated category, then each is a triangulated subcategory.

Proof. Let us prove this for $\mathcal{D}_H^+$. It is clear that it is preserved under [1] and [−1]. If (X, Y, Z, f, g, h) is a distinguished triangle of $\mathcal{D}$ and $H(X[n]) = H(Y[n]) = 0$ for all $n \ll 0$, then also $H(Z[n]) = 0$ for all $n \ll 0$ by the long exact sequence (0148). Hence we may apply Lemma 05QX to see that $\mathcal{D}_H^+$ is a pre-triangulated subcategory (respectively a triangulated subcategory if $\mathcal{D}$ is a triangulated category). The

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assertion of being saturated follows from

$$\begin{aligned} H((X \oplus Y)[n]) = 0 &\Rightarrow H(X[n] \oplus Y[n]) = 0 \\ &\Rightarrow H(X[n]) \oplus H(Y[n]) = 0 \\ &\Rightarrow H(X[n]) = H(Y[n]) = 0 \end{aligned}$$

for all $n \in \mathbf{Z}$. □